

## DITO Model Development

### Sample details:

The model was developed on 75% and validated on 25% of the sample.

|                           |                            |
|---------------------------|----------------------------|
| <b>Observation Period</b> | January 2023 – August 2025 |
| <b>Development (75%)</b>  | 10,147                     |
| <b>Validation (25%)</b>   | 3,383                      |

### Good/Bad Definition:

| GB Flag Value         | GB Flag Definition                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------|
| <b>Outcome Period</b> | 1 month                                                                                     |
| <b>Bad</b>            | DPD30+ - Clients who delayed instalment with more than 30 days after first month            |
| <b>Good</b>           | DPD30<30 - Clients who have not delayed instalment with more than 30 days after first month |

| Development Sample     |       |
|------------------------|-------|
| <b>Goods</b>           | 8,461 |
| <b>Bads</b>            | 1,686 |
| <b>Sample Bad Rate</b> | 16.6% |
| Validation Sample      |       |
| <b>Goods</b>           | 2,830 |
| <b>Bads</b>            | 553   |
| <b>Sample Bad Rate</b> | 16.4% |

### Bespoke model - Score Distribution -Train:

Gini: 32.4%

| Band | Score Range | Bad | Good | All  | Bad Rate |
|------|-------------|-----|------|------|----------|
| 1    | 300-448     | 334 | 637  | 971  | 34.4%    |
| 2    | 449-465     | 248 | 777  | 1025 | 24.2%    |
| 3    | 466-473     | 220 | 746  | 966  | 22.8%    |
| 4    | 474-480     | 190 | 831  | 1021 | 18.6%    |
| 5    | 481-487     | 149 | 782  | 931  | 16.0%    |
| 6    | 488-494     | 143 | 921  | 1064 | 13.4%    |
| 7    | 495-501     | 133 | 857  | 990  | 13.4%    |
| 8    | 502-509     | 124 | 941  | 1065 | 11.6%    |
| 9    | 510-520     | 80  | 975  | 1055 | 7.6%     |
| 10   | 521-600     | 65  | 994  | 1059 | 6.1%     |

### Bespoke model - Score Distribution - Test:

Gini: 24.5%

| Band | Score Range | Bad | Good | All | Bad Rate |
|------|-------------|-----|------|-----|----------|
| 1    | 300-448     | 90  | 216  | 306 | 29.4%    |
| 2    | 449-465     | 71  | 255  | 326 | 21.8%    |
| 3    | 466-473     | 67  | 247  | 314 | 21.3%    |
| 4    | 474-480     | 62  | 274  | 336 | 18.5%    |
| 5    | 481-487     | 55  | 295  | 350 | 15.7%    |
| 6    | 488-494     | 57  | 310  | 367 | 15.5%    |
| 7    | 495-501     | 52  | 299  | 351 | 14.8%    |
| 8    | 502-509     | 40  | 312  | 352 | 11.4%    |
| 9    | 510-520     | 35  | 308  | 343 | 10.2%    |
| 10   | 521-600     | 24  | 314  | 338 | 7.1%     |

## Bespoke model - Score Distribution - Train + Test:

| Band | Score Range | Bad | Good | All  | Bad Rate |
|------|-------------|-----|------|------|----------|
| 1    | 300-448     | 424 | 853  | 1277 | 33.2%    |
| 2    | 449-465     | 319 | 1032 | 1351 | 23.6%    |
| 3    | 466-473     | 287 | 993  | 1280 | 22.4%    |
| 4    | 474-480     | 252 | 1105 | 1357 | 18.6%    |
| 5    | 481-487     | 204 | 1077 | 1281 | 15.9%    |
| 6    | 488-494     | 200 | 1231 | 1431 | 14.0%    |
| 7    | 495-501     | 185 | 1156 | 1341 | 13.8%    |
| 8    | 502-509     | 164 | 1253 | 1417 | 11.6%    |
| 9    | 510-520     | 115 | 1283 | 1398 | 8.2%     |
| 10   | 521-600     | 89  | 1308 | 1397 | 6.4%     |

## Model Development Stages:

- **Data audit and preparation**  
*Cleaning the Telco data from data with lower quality (not populated records, etc.). Deriving a set of features based on the raw Telco data.*
- **Target definition**  
*Defining the target definition. This step requires collaboration with the client's Risk team*
- **Sample selection**  
*Choosing the most appropriate sample window based on the Target definition, seasonality, past marketing campaigns, etc.*
- **Features selection**  
*Creating a short list with features to be tested in the model development*
- **Model Training on 75%**  
*Building a model using only 75% of the Accepted accounts – Probability of Default*
- **Model Validation on 25%**  
*Validating the model using the remaining 25% of the Accepted accounts – Probability of Default*

## Machine learning techniques used in Model Development:

- FinScore initially develops a Logistic Regression model and use its predictive power as a base. Then attempts various ML algorithms until coming up with the best possible result
- ML techniques that are most frequently attempted:
  - XGBoost
  - Random Forrest (usually when the sample size it small)
  - RBM Neural Network
- **Current ML technique that outperformed the rest and was finally selected was XGBoost**

## Post-paid customers

| Score | Band | Score | Risk Level |               |
|-------|------|-------|------------|---------------|
| 21    |      | 880   | Highest    | Dito Postpaid |
| 22    |      | 920   | Medium     |               |
| 23    |      | 960   | Lower      |               |

### Telco Data groups:

FinScore receives more than 400 raw features from Dito for every single cellphone number. The following groups of Telco data are included:

- **Voice usage** – outgoing calls (minutes, number of calls, duration, etc.)
- **SMS usage** - outgoing SMSs (number, time of the day, etc.)
- **Mobile Data usage** – used mobile data (megabytes, frequency, etc.)
- **Top-up patterns** – recharges (amounts, number of recharges, frequency, etc.)
- **Sim card data** – activation date, last usage, etc.)
- **Behavior patterns** – sending load to other individuals, topping up from different sources, etc.)
- **Spending patterns** – amounts, packages/bundles, frequencies, etc.)